B. Tech 3rd Sem (IBS) Assignment-1
Deadline: Next Class

A. Consider the following mRNA:
AGU CUC UGU CUC CAU UUG AAG AAG GGG AAG GGG
a) Indicate the amino acid sequence which would be coded using genetic code table. Is it a full length protein sequence?
b) You obtain mutations which consist of additions or deletions of one nucleotide. If we insert G between the third and fourth nucleotide, and we eliminate the 10th nucleotide from the right (it is a G), will the consequence remain the same?

B. Go to Google→BLAST→CLICK ON BLAST: Basic Local Alignment Search Tool→ Nucleotide BLAST→Paste the following nucleotide **sequence (A)** in BLAST nucleotide window

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ATGTCTTCTGGCAATTTCATATAAGTATTTTTTCAAAAATGTCTCTTCTGTCAACCCCACGCCTTTG
GCACAATGAAGTGGGTAACCTTTATTTCCCTTCTTTTTCTCTTTAGCTCGGCTTATTCCAGGGGTGTGTT
TCGTCGAGATGCACACAAGAGTGAGGTTGCTCATCGGTTTAAAGATTTGGGAGAAGAAAATTTCAAAGCC
TTGGTGTTGATTGCCTTTGCTCAGTATCTTCAGCAGTGTCCATTTGAAGATCATGTAAAATTAGTGAATG
AAGTAACTGAATTTGCAAAAACATGTGTAGCTGATGAGTCAGCTGAAAATTGTGACAAATCACTTCATAC
CCTTTTTGGAGACAAATTATGCACAGTTGCAACTCTTCGTGAAACCTATGGTGAAATGGCTGACTGCTGT
GCAAAACAAGAACCTGAGAGAAATGAATGCTTCTTGCAACACAAAGATGACAACCCAAACCTCCCCCGAT
TGGTGAGACCAGAGGTTGATGTGATGTGCACTGCTTTTCATGACAATGAAGAGACATTTTTGAAAAAATA
CTTATATGAAATTGCCAGAAGACATCCTTACTTTTATGCCCCGGAACTCCTTTTCTTTGCTAAAAGGTAT
AAAGCTGCTTTTACAGAATGTTGCCAAGCTGCTGATAAAGCTGCCTGCCTGTTGCCAAAGCTCGATGAAC
TTCGGGATGAAGGGAAGGCTTCGTCTGCCAAACAGAGACTCAAATGTGCCAGTCTCCAAAAATTTGGAGA
AAGAGCTTTCAAAGCATGGGCAGTGGCTCGCCTGAGCCAGAGATTTCCCAAAGCTGAGTTTGCAGAAGTT
TCCAAGTTAGTGACAGATCTTACCAAAGTCCACACGGAATGCTGCCATGGAGATCTGCTTGAATGTGCTG
ATGACAGGGCGGACCTTGCCAAGTATATCTGTGAAAATCAGGATTCGATCTCCAGTAAACTGAAGGAATG
CTGTGAAAAACCTCTGTTGGAAAAATCCCACTGCATTGCCGAAGTGGAAAATGATGAGATGCCTGCTGAC
TTGCCTTCATTAGCTGCTGATTTTGTTGAAAGTAAGGATGTTTGCAAAAACTATGCTGAGGCAAAGGATG
TCTTCCTGGGCATGTTTTTGTATGAATATGCAAGAAGGCATCCTGATTACTCTGTCGTGCTGCTGCTGAG
ACTTGCCAAGACATATGAAACCACTCTAGAGAAGTGCTGTGCCGCTGCAGATCCTCATGAATGCTATGCC
AAAGTGTTCGATGAATTTAAACCTCTTGTGGAAGAGCCTCAGAATTTAATCAAACAAAACTGTGAGCTTT
TTAAGCAGCTTGGAGAGTACAAATTCCAGAATGCGCTATTAGTTCGTTACACCAAGAAAGTACCCCAAGT
GTCAACTCCAACTCTTGTAGAGGTCTCAAGAAACCTAGGAAAAGTGGGCAGCAAATGTTGTAAACATCCT
GAAGCAAAAAGAATGCCCTGTGCAGAAGACTATCTATCCGTGGTCCTGAACCAGTTATGTGTGTTGCATG
AGAAAACGCCAGTAAGTGACAGAGTCACAAAATGCTGCACAGAGTCCTTGGTGAACAGGCGACCATGCTT
TTCAGCTCTGGAAGTCGATGAAACATACGTTCCCAAAGAGTTTAATGCTGAAACATTCACCTTCCATGCA
GATATATGCACACTTTCTGAGAAGGAGAGACAAATCAAGAAACAAACTGCACTTGTTGAGCTTGTGAAAC
ACAAGCCCAAGGCAACAAAAGAGCAACTGAAAGCTGTTATGGATGATTTCGCAGCTTTTGTAGAGAAGTG
CTGCAAGGCTGACGATAAGGAGACCTGCTTTGCCGAGGAGGGTAAAAAACTTGTTGCTGCAAGTCAAGCT
GCCTTAGGCTTATAACATCTACATTTAAAAGCATCTCAGCCTACCATGAGAATAAGAGAAAGAAAATGAA
GATCAAAAGCTTATTCATCTGTTTTCTTTTTCGTTGGTGTAAAGCCAACACCCTGTCTAAAAAACATAAA
TTTCTTTAATCATTTTGCCTCTTTTCTCTGTGCTTCAATTAATAAAAAATGGAAAGAATCTAATAGAGTG
GTACAGCACTGTTATTTTTCAAAGATGTGTTGCTATCCTGAAAATTCTGTAGGTTCTGTGGAAGTTCCAG
TGTTCTCTCTTATTCCACTTCGGTAGAGGATTTCTAGTTTCTGTGGGCTAATTAAATAAATCACTAATAC
TCTTCTAA
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1) **What is BLAST?**
2) **What is the length of the query sequence (A)?**
3) **Identify the sequence (name, organism, accession id).**

4) **Suppose for your identified sequence you got an E value zero for 100% identity- What does it mean?**
5) **Could you find out the function of the protein?**

Now, Go to Google and type PUBMED. Click on pubmed to reach the site @ *www.ncbi.nlm.nih.gov/pubmed.* Type **Bovine Serum Albumin/BSA** (which is serum albumin from cattle) in the window and scroll down pubmed window to specify nucleotide. You will find "Bos taurus mRNA for **bovine serum albumin**", click on that. You will see the **sequence (B)**

6) How long is the nucleotide sequence (B)?

Now, click on FASTA (a text-based format for representing nucleotide sequences or peptide sequences) and you will see the sequence B which is the nucleotide sequence of **Bovine Serum Albumin (BSA).** Copy the sequence (B).

Go to Google→BLAST→click on BLAST: Basic Local Alignment Search Tool→Nucleotide BLAST→**Select Align two or more sequences**→paste nucleotide sequence (A) and the nucleotide sequence (B) in TWO BLAST nucleotide windows →Run BLAST.

7) What % of identity you found between sequence A and B. What will be your conclusion?

Again go to Google and type PUBMED. Click on pubmed to reach the site @ *www.ncbi.nlm.nih.gov/pubmed.* Type Bovine Serum Albumin in the window and scroll down pubmed window to specify Protein. You will find "bovine serum albumin [Bos taurus]", click on that. You will see the protein sequence (B). In the same page, at the right hand site you will see "Protein 3D Structure" and there click on "see all 5 structures". You will find two entries of "Crystal Structure Of Bovine Serum Albumin [Transport Protein]" among five structures. Click on both.

8) How big is the amino acid sequence of BSA?

9) What are the protein data base ids (PDB ID) for the two submitted protein structures for the same BSA protein? In which journals both the submitted sequences were published? Give details of the publication (Journal name, volume, page no, year). What are the titles of those two papers? Write down the na mes of the authors.

10) Download PyMOL Molecular Graphic System in your computer. Load both PDB IDs one after another (Plugin→PDB loader service). Show two structures as Cartoon (hiding side chains and water); one colored as red and other one colored as Yellow. Superimpose them. Are you convinced that both are the same structures? Take a snap shot and submit.